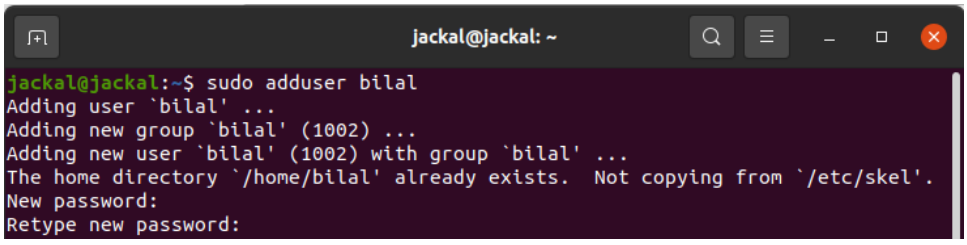
Lab 2 (b)

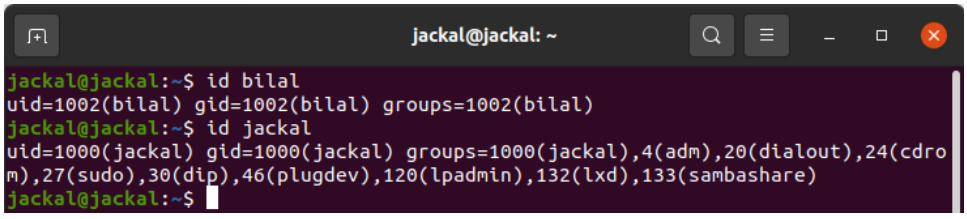
Changing Ownership:

Creating user: To understand ownership we need to make another user by executing the following command.

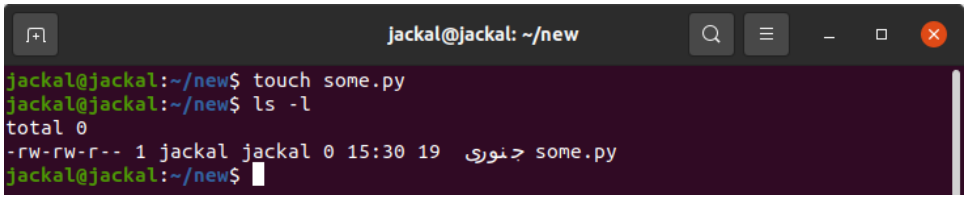
Command: sudo adduser <username>.



Checking ID of the user: Use the id <username> command to print the ID assigned to user as below. As we can see bilal has 1002 id and jackal (main user) have 1000 id and other supplementary ids.

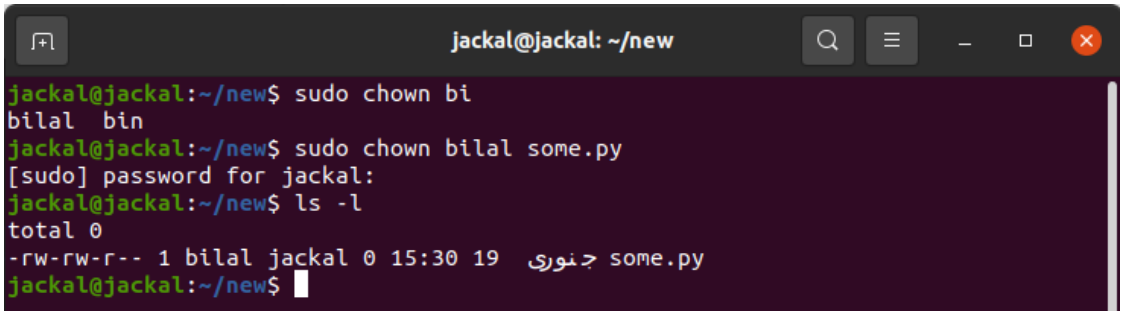


Changing ownership of a file some.py: As we have now multiple users, we can change the ownership of the file to another user as below. Let's create a new file and check permissions and ownership.

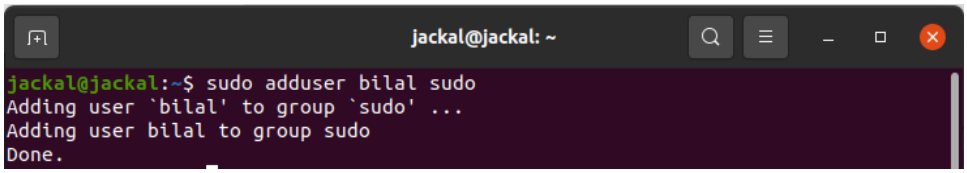


And now change the ownership using following command.

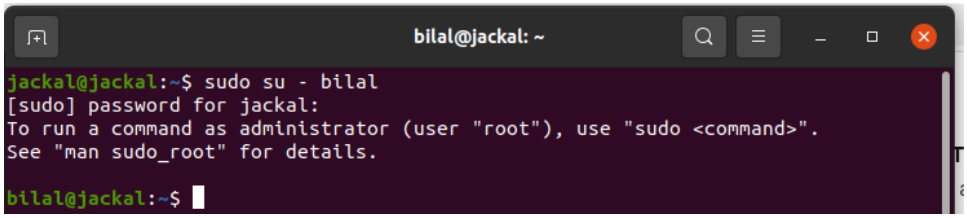
Command: sudo chown <user> <file>.



Sudo Group: We can add a user in sudo group as below.



Login to user: We can login to another user as below.



Changing Permissions:

Command: In Linux we use **chmod** command to change the permission of any file for groups. We can change the permission using two ways; **symbolic and octal mode.**

Symbolic mode: This method uses operator, letters and group type or reference. Have a look at operators in Table 1, letters in Table 2 and Reference in Table 3.

Table 1.

Operator	Work
“+”	Add Permission
“-”	Remove Permission
“=”	Assign or set the permission

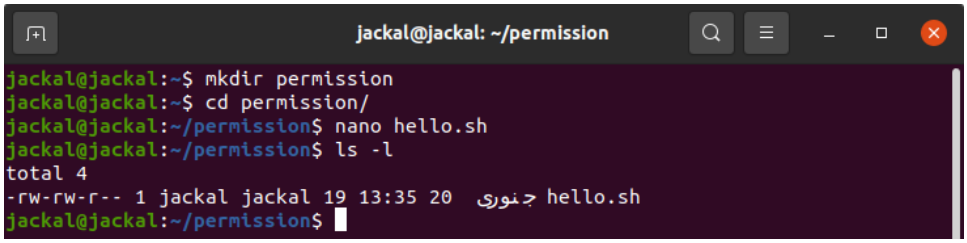
Table 2.

Letter	Permission Type
"r"	Read
"w"	Write
"x"	Execute

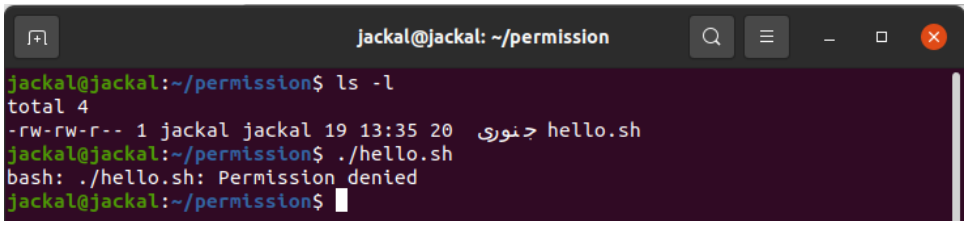
Table 3.

Reference	Group Type
"u"	Owner
"g"	Group(s)
"o"	Others
"a"	All

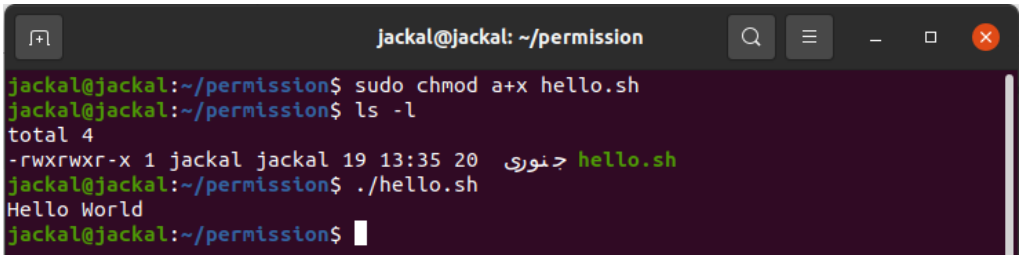
Usage: Let's create a simple shell script with name hello.sh, which only has one line => echo "Hello World", save it and check its permissions as below shown.



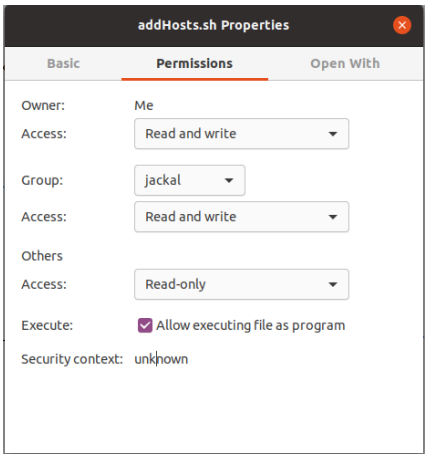
Execution Permission: We won't be able to run this script as it doesn't have execution permission as shown below.



Change Permission: Let's give all groups permission for the execution, check the permission and run it as below shown.



Using GUI: We can also change the permissions of file using GUI of Ubuntu by going to properties of file and then permissions as below.



Octal Mode: Uses three-digit octal number to set permissions. The first digit sets the permission for the owner, second digit for group and third digit for others. See the Table 4 for usage of this method.

Table 4.

Number	Permission(s)
0	---
1	--x
2	-w-
3	-wx
4	r--
5	r-x
6	rw-
7	rwx

Usage: Let's change the permissions of hello.sh created earlier as below using octal method. We will give all permissions to the owner, read permission to the group and no permission to others.

Compile C code:

Writing Code: Below is simple C code. Create file using nano and add name this as hello.c

```
#include <stdio.h>

int main() {
    // This is a simple C program
    printf("Hello, World!\n");
    return 0;
}
```

Compile: Compile it using gcc command and produce its executable file.

Command: gcc hello.c -o hello

Execution Command 1: bash hello.c
Execution Command 2: ./hello.c

Overall Procedure and Output: